17 Building Automation

By Ken Sinclair

Topic Highlights

History of the Evolution to Direct Digital Control (DDC)
Example of the Application of DDC
Open Protocols Use in Building Automation
How to Specify Building Automation Systems
Future of Web Services in Building Automation
Web-Based Facilities Operations Guide

17.1 Introduction & Overview

This chapter is designed to provide insight into the industry that automates large buildings. Each large building has a custom-designed heating, ventilating, and cooling (HVAC) air conditioning system to which automated controls are applied. Access control, security, fire, life safety, lighting control, and other building systems are also automated as part of the building automation system. These systems are viewed by building management on-site and off-site with standard computer technology. In the past, this was done by proprietary communication systems using telephone modems, but now the trend is to browser-based presentation using Internet access with open communication standards.

The names given to building automation systems are varied:

- Building Automation Systems or BAS
- Building Management Systems or BMS
- Facility Management Systems or FMS
- Energy Management Systems or EMS
- Energy Management Control Systems or EMCS
- Client Comfort System or CCS

plus others, based on building-based functions such as lighting control, fire/life safety, security, video, digital signage, etc.

Basic climate control systems (building HVAC) vary widely, depending on owner and building requirements and the mechanical engineer's design. To provide insight into the complete process, this chapter includes an example of how the instrument design for a typical air handing system is laid out. The building zoning is determined by the design engineer and owner of the building and is based on cost versus comfort. Thermal zones usually reflect the internal thermal barriers of the building.

Most controls are based on simple feedback of space temperature. However, the complexity comes from optimizing the interaction of the central equipment—such as the chillers, boilers, pumps, and air-handling equipment—with the great number of individual zones that make up the HVAC system.

Solar control is sometimes used to operate window shades, but most North American building designs depend on brute strength to overcome heating and cooling requirements. The green building movement is slowly changing this type of thinking, encouraging passive designs and interactive control of the building envelope. Outdoor air is used when possible for free cooling, but many buildings have only minimum outside air to provide for ventilation. Temperature is generally sensed with low-cost thermistors.

The energy management control for a building can be as simple as time clock control or as complex as Gridwise, which provides connections based on interactive Web information that allow the building to be a controllable load on the nation's electrical grid. In smaller buildings and homes, the controls are often packaged and provided with the heating or cooling equipment.

17.2 History of the Evolution to Direct Digital Control or DDC

A brief history of control evolution provides insight to the development of direct digital control, or DDC, an industry foundation. The automated buildings evolution followed the growth of the pneumatic control industry for a century, seeing the gradual introduction of pneumatic transmission, followed by electric and electronic control. In the mid-1960s, electronic control evolved to multiplex control systems that evolved into the first computerized systems using head-end computers. This quickly gave way to minicomputers, then ended with microcomputers and the DDC revolution. In the late 1970s and early 80s, the use of DDC exploded, greatly expanding the scope of the traditional building automation control market while displacing the traditional pneumatic control industry. The automated building control market growth curve went almost vertical. The decade between mid-1980 and mid-1990 saw the growth of many new control companies able to bring low cost, high functionality, proprietary protocol control systems rapidly to the marketplace. By the mid-1990s, the cost of DDC was much lower than the cost of pneumatics. This rapidly fueled the replacement of pneumatic controls and the expansion of traditional and non-traditional automation markets.

During the mid-1990s, the BACnet movement, "one of the first open communication protocols," began to gain momentum, and a few systems were built around the evolving standard. About the same time, the industry started to follow the revolution that was occurring in other industries with products based on the Echelon chip. Standard protocols became something the industry wanted to achieve. Slowly, but surely, the number of proprietary communication standards were reduced and replaced by open-standard communication protocols.

The next revolution to hit was the Internet/intranet and browser-based presentations of everything. This forced all companies to look hard at what the information technology (IT) industry was doing to standardize presentation models. Some companies had built their new systems around IT thinking and were able to rapidly follow, and even lead, this new trend. Once DDC concepts were embedded in browser-based presentations, their convergence fueled rapid evolution with wireless cell phones, personal digital assistant (PDA) interfaces, e-mailed alarms, etc. With this followed the rapid evolution that further explored Internet capabilities and increased the understanding of the complete power of the newly inherited tools from the Web environment; with this, building automation functionality exploded. Figure 17-1 provides a visual portrayal of all these events.

17.2.1 Example of the Application of DDC

An example of the points list for a basic DDC for an air handling system—shown in Table 17-1 on the following page, with accompanying Figure 17-2 following—is reprinted here with permission from its source, BC Buildings Corp. (BCBC), Victoria, British Columbia, Canada.

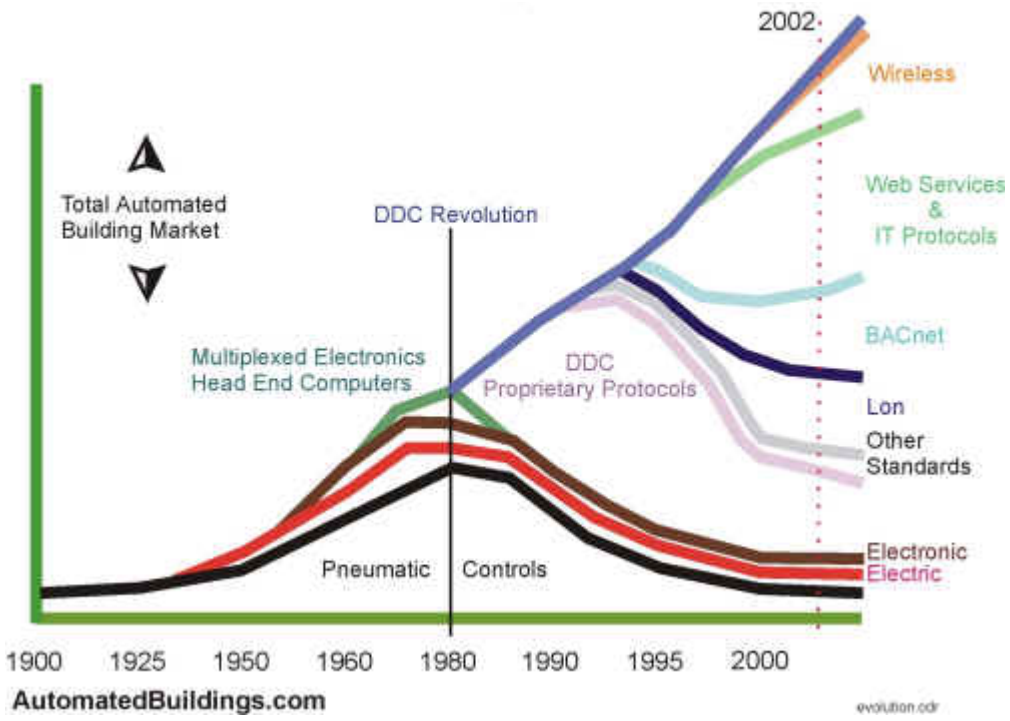


Figure 17-1: Automated Buildings Evolution

The *BCBC Client Comfort System Design Manual* is primarily intended to instruct and assist designers who are specifying computerized building controls for BC Buildings Corp. It starts with an introduction that describes how to use the manual for the procurement process.

Once you conceptually understand how this is done, you will see how automation is applied to other building components. Sensing and control points are provided for each air system, plus all other major pieces of equipment provided in the mechanical building design. Both hardware and software points are specified.

The points list provides each hardware and software point with a unique number, shown on Figure 17-2. The point type is indexed in a specification which provides a complete detailed specification for each end sensing and control device as well as all software points.

The graphic of how this system should be represented on the computer screen is part of the specification and shows the exact location of each of the required sensing points and the actuated devices. Software points, which are detailed in the specification, are also shown. The actual location on the graphic where the operator would click to get trend information, time schedules, access to control programs, etc., is also shown.

The graphic is followed by the control language, which shows the exact relationship between the real and virtual points in the automation system presented in a basic If, Then, Else, control language.

This is an example of one air system. In an extremely large building, there could be hundreds of air systems, typically one per floor in high-rises, plus thousands of individual terminal zones—each with their own DDC control. In addition, central equipment such as chillers and boilers to provide heating or cooling all have their own control systems. These DDC control points are all networked together to allow interaction of all points. This is done in the control language and the presentation to the operator is via a browser-based interface.

Table 17-1: Example Points List for a Basic DDC Air Handling System (Source: BC Buildings Corp. Reprinted with permission.)

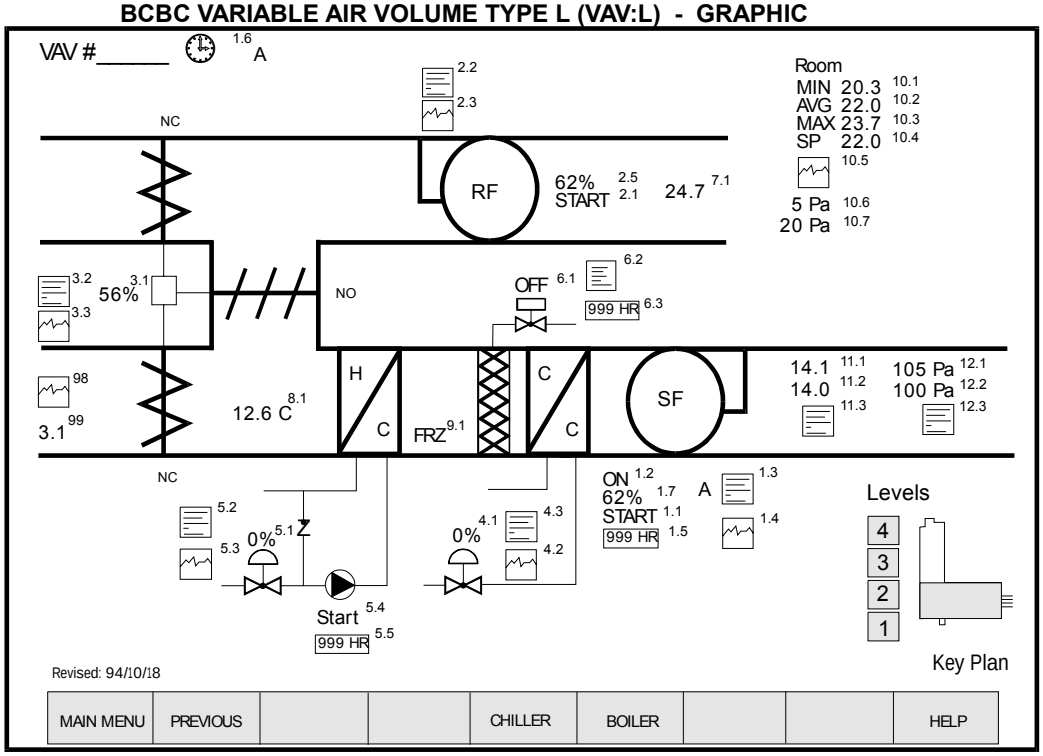
Project Number:		Project Name:									
Graphic Logic Location		Point Name Mnemonic		Point	Hardware Point				Virtual	Notes See Page #	Note
	Point Description	System	Point	Type	DO	DI	AO	AI	Point	Comments	#
1.1	Supply Fan Start/Stop		SF	CR1	1						
1.2	Supply Fan Status		SF_S	V					1		
1.3	Supply Fan Program		SF_PG	PG					1		
1.4	Supply Fan Trendlog		SF_TL	TL					1	SF,SF_ASD,SAT,SAP	
1.5	Supply Fan Run Hours		SF_TZ	TZ					1		
1.6	Weekly Schedule		WS	WS					1		
1.7	Supply Fan ASD		SF_ASD	DDC			1				
2.1	Return Fan Start/Stop		RF	CR1	1						
2.2	Return Fan Program		RF_PG	PG					1		
2.3	Return Fan Trendlog		RF_TL	TL					1	RF,RAT,BSP,ASD	
2.5	Return Fan ASD		RF_ASD	DDC			1				
3.1	Mixed Air Damper		MAD	DA2			1				
	Mixed Air Damper Controller		MAD_CO	CO					1	use SAT:SAT_SP	
	Mixed Air Damper Minimum		MAD_MIN	V					1		
	Mixed Air Damper Ramp		RAMP	V					1		
3.2	Mixed Air Program		MAD_PG	PG					1		
3.3	Mixed Air Trendlog		MAD_TL	TL					1	SF_S,FRZ,MAD,MAT	
	Cooling Mode		CLG_MODE	V					1		
4.1	Cooling Coil Valve		CCV	CV3			1				
	Cooling Coil Controller		CCV_CO	CO					1	use SAT:SAT_SP	
4.2	Cooling Coil Trendlog		CLG_TL	TL					1	CCV,SAT_SP,SAT,EC	
4.3	Cooling Coil Program		CCV_PG	PG					1		
	Heating Mode		HTG_MODE	V					1		
5.1	Heating Coil Valve		HCV	CV3			1				
	Heating Coil Controller		HCV_CO	CO					1	use SAT:SP = 12	
5.2	Heating Program		HTG_PG	PG					1		
5.3	Heating Coil Valve Trendlog		HCV_TL	TL					1	HCV,SAT_SP,SAT,HCP	
5.4	Heating Coil Pump		HCP	CR1	1						
5.5	Heating Coil Pump Run Hours		HCP_TZ	TZ					1		
6.1	Evaporative Cooling		EC	CR1	1						
6.2	Evaporative Cooling Program		EC_PG	PG					1		
6.3	Evaporative Cooling Run Hours		EC_TZ	TZ					1		
7.1	Return Air Temperature		RAT	DTS2				1			
8.1	Mixed Air Temperature		MAT	DTS1				1			
9.1	Freeze Control		FRZ	FRZ		1					
10.1	Room Temperature Minimum		RT_MIN	V					1		
10.2	Room Temperature Average		RT_AVG	V					1		
10.3	Room Temperature Maximum		RT_MAX	V					1		
10.4	Room Temperature Setpoint		RT_SP	V					1		
10.5	Room Temperature Trendlog		RT_TL	TL					1	SAT,MIN,AVG,MAX	
10.6	Building Pressure		BSP	DPS				1			
10.7	Building Pressure Setpoint		BSP_SP	V					1		
11.1	Supply Air Temperature		SAT	DTS2				1			
11.2	Supply Air Temp Setpoint		SAT_SP	V					1		
11.3	Supply Air Temp Program		SAT_PG	PG					1		
12.1	Supply Air Pressure		SAP	DPS				1			
12.2	Supply Air Pressure Setpoint		SAP_SP	V					1		
12.3	Supply Air Pressure Program		SAP_PG	PG					1		
	Supply Air Pressure Controller		SAP_CO	CO					1	use SAP:SAP_SP	
	Total				4	1	5	5	34		

17.3 Open Protocols used in Building Automation

In the past, proprietary protocols were used to network DDC points. However, today it is more likely open protocols such as BACnet, LonWorks, and/or TCP/IP, oBIX , and Niagara standards would be used.

BACnet is an acronym for a data communication protocol for Building Automation and Control Networks. The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) maintains a BACnet website. See: http://www.bacnet.org/.

BCBC VARIABLE AIR VOLUME TYPE L (VAV:L) - GRAPHIC



GENERIC START UP LOGIC

1.3 **Supply Fan (SF_PG)**
IF RF ON-FOR 30 SEC THEN
START SF
ELSE STOP SF
IF SAP > 30 THEN SF_S = ON
ELSE SF_S = OFF
[Supply Fan ASD]
IF SF ON THEN
SF_ASD = SAP_CO
ELSE SF_ASD = 0

2.2 **Return Fan (RF_PG)**
IF WS ON OR RT_AVG < 14 THEN
START RF
IF WS OFF AND RT_AVG > 16 THEN
STOP RF
[Return Fan ASD]
BSP_SP = _____ *DEFAULT = 20 Pa*
IF RF ON THEN
RF_ASD = SF_ASD - (BSP_SP - BSP)
ELSE RF_ASD = 0

3.2 **Mixed Air Damper (MAD_PG)**
MAD_MIN = _____ *DEFAULT = 30%*
DO EVERY 3 S RAMP = RAMP + 1%
IF SF OFF THEN
RAMP = 0%
IF RAMP > 100% THEN
RAMP = 100% END DO
IF SF_S ON AND WS ON AND FRZ OFF THEN
IF CLG_MODE ON OR HTG_MODE ON THEN
IF RAT < OAT OR HTG_MODE ON THEN
MAD = MAD_MIN
ELSE MAD = 100%
ELSE
MAD = MAX(MAD_MIN, MAD_CO)
MAD = MIN(MAD, RAMP)
ELSE MAD = 0

4.3 **Cooling (CLG_PG)**
IF SF_S ON THEN
IF RT_AVG > RT_SP + 1.3 OR
RT_MAX > RT_SP + 2 THEN
CLG_MODE = ON, CCV = CCV_CO
IF RT_AVG < RT_SP + 0.8 AND
RT_MAX < RT_SP + 1.5 THEN
CLG_MODE = OFF, CCV = 0
ELSE
CLG_MODE = OFF, CCV = 0

5.2 **Heating (HTG_PG)**
IF SAT < 10 THEN
HTG_MODE = ON, HCV = HCV_CO
IF SAT > 14 THEN
HTG_MODE = OFF, HCV = 0
[Heating Coil Pump]
IF HCV > 15 OR MAT < 3 THEN
START HCP
IF HCV < 2 AND MAT > 7 THEN
STOP HCP

6.2 **Evaporative Cooling (EC_PG)**
IF SF_S ON AND CLG_MODE OFF THEN
IF RT_AVG > RT_SP + 0.5 OR
RT_HIGH > RT_SP + 1 THEN
START EC
IF RT_AVG < RT_SP + 0.1 THEN
STOP EC
ELSE STOP EC

11.3 **Supply Air Temp (SAT_PG)**
RT_SP = ____ *DEFAULT = 22*
SAT_SP = 18 - ((RT_AVG - RT_SP) * 4)

12.3 **Supply Air Pressure (SAP_PG)**
SAP_SP = ____ *DEFAULT = 100 Pa*

5_VL1.CDR Updated: 94/10/31 by KWS Page _____

Figure 17-2: Unique Number Provided for Each Hardware and Software Point (Source: BC Buildings Corp. Reprinted with permission.)

LonMark International is a global membership organization created to promote and advance the business of efficient and effective integration of open, multi-vendor control systems utilizing ANSI/EIA/CEA 709.1 and related standards. See: http://www.lonmark.org/.

TCP/IP is the acronym for Transmission Control Protocol/Internet Protocol, the suite of communications protocols used to connect hosts on the Internet.

oBIX stands for Open Building Information Xchange, an industry wide initiative to define XML- and Web Services-based mechanisms to present building systems-related information on TCP/IP networks such as the Internet. See: http://www.obix.org/.

Niagara is a different way to address the challenge of creating true interoperable and open systems. Niagara takes data from diverse systems and "morphs" all of the data into uniform software "components." It's a comprehensive software framework designed to address the challenges of integrating diverse smart devices into unified systems. These components can then easily be assembled into applications—for example, dynamic displays, control sequences (even across different systems and devices), alarms, schedules, reports, etc. Niagara makes it possible to expose all this data to the enterprise in a unified IT-friendly way. Niagara combines the capabilities of network management, protocol conversion, distributed control, and the user interface into a single software solution that can operate on a wide range of hardware platforms from very small to the very large. It can be used on PCs, on servers and in small, embedded controllers. See: http://www.tridium.com/.

The art of the automated buildings industry is connecting and creating the correct relationships of these thousands of DDC points, while providing communication to a Web browser with open protocols. Rapid evolution in the automated buildings industry has been fuelled with the standardization of communication protocols.

Figure 17-3 is from the Reliable Controls Web site at http://www.reliablecontrols.com/, and is reprinted here with permission. It provides a good example of how an open protocol operating system using BACnet protocol takes the DDC points and creates a relationship for each point in a control language (BASIC Code) and displays them on a standard internet browser.

17.4 How to Specify Building Automation Systems

What is CtrlSpecBuilder?

CtrlSpecBuilder™ is a free online productivity tool for preparing bid specifications for HVAC control systems. CtrlSpecBuilder can prepare BACnet specifications or specs that allow other protocols. You can view the spec online and download it as a Microsoft Word file.

CtrlSpecBuilder provides tool for preparing quicker, easier specifications

A new Website has been launched providing HVAC engineers with a new productivity tool. Previously, engineers wrote specifications from scratch, cut and pasted from previous projects, or relied on proprietary vendor "canned" specifications. CtrlSpecBuilder is a free, online tool that prepares non-proprietary, custom specifications. It follows ASHRAE Guideline 13-2000, specifying DDC systems, and CSI MasterFormat for section 15900. This tool also generates point lists and sequences of control for all HVAC equipment in the project, and can provide specifications in U.S. or metric units.

CtrlSpecBuilder can be accessed at http://www.ctrlspecbuilder.com/.

17.5 Future of Web Services in Building Automation

The following is an excerpt from an article published January 2002 on AutomatedBuildings.com's Web site. The article, titled *Information Model: The Key to Integration* at http://www.automatedbuildings.com/news/jan02/art/alc/alc.htm, has been widely read and is a cornerstone for Web-based evolu-

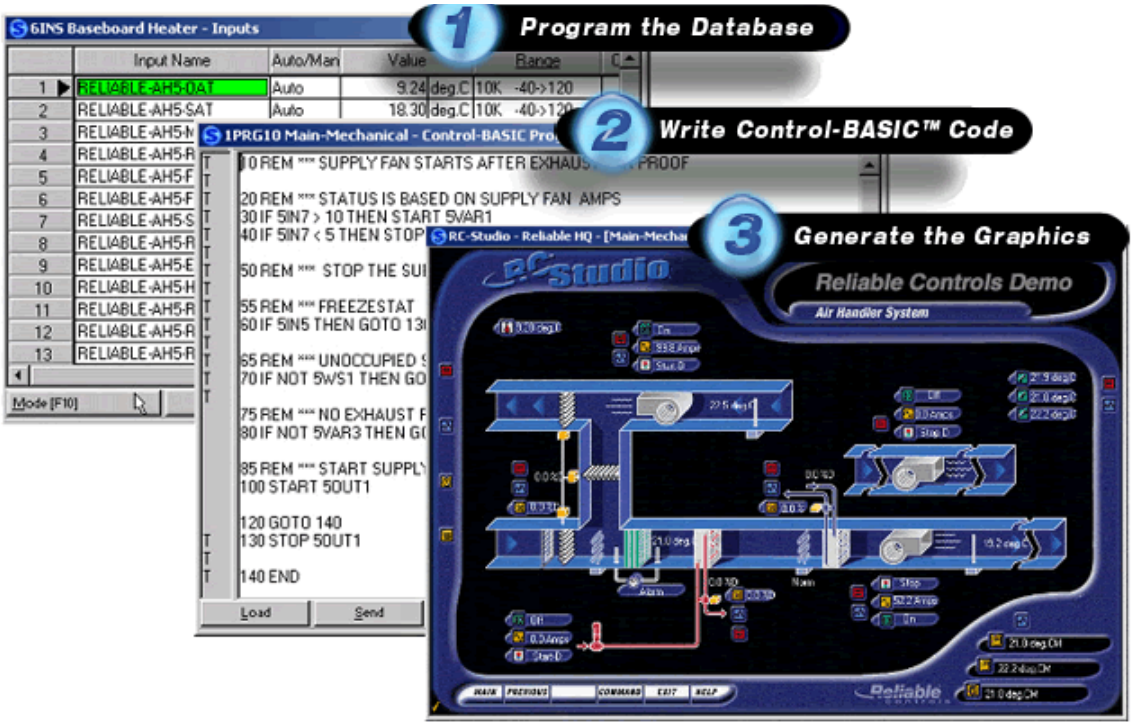


Figure 17-3: Display Showing How BACnet Protocol Is Applied
(Source: Reliable Controls Corp., Victoria, British Columbia, Canada. Reprinted with permission.)

tion. Its authors are Eric Craton, head of product development, and Dave Robin, head of software development, at Automated Logic Corp., Atlanta, Ga.

Let's look at the four trends in terms of Web services:

1. **Content is becoming dynamic** - A Web service has to be able to combine content from many different sources. That may include furniture inventories, maintenance schedules and work orders, energy consumption and forecasts, as well as traditional building automation information.

2. **Bandwidth is getting cheaper** - A Web service can now deliver types of content (streaming video or audio, for example) unthinkable a few years ago. As bandwidth continues to grow, Web services must adapt to new content types.

3. **Storage is getting cheaper** - A Web service must be able to deal with massive amounts of data intelligently. That means using technologies such as database replication, LDAP directories, caches, and load balancing software to make sure that scalability isn't an issue.

4. **Enterprise computing is becoming more important** - A Web service can't require that users run a traditional browser on some version of Windows. Web services have to serve all sorts of devices, platforms, and browser types, delivering content over a wide variety of connection types for a wide variety of purposes.

17.6 Web-Based Facilities Operations Guide

In August of 2002, I prepared a supplement for *Engineered Systems Magazine*, called *A Guide to Web-Based Facilities Operations.* See: http://www.automatedbuildings.com/news/aug02/articles/ksin/ksin.htm. Here's an excerpt:

"Doing more with less by using Web-based anywhere information to amplify your existing building operational resources.

"The reality of today and tomorrow's economy is that we will be doing more with less to effectively manage our buildings. Our saviour is that technology in the form of Web-based everything is now providing us a path to improved communications, while simplifying the movement of complex building information to the evolving Building Operations Specialist.

"The rapid movement of the building automation industry towards Web-based allows us to interweave critical dynamic building information into a browser-based anywhere presentation. This allows us to concentrate and amplify our existing building operation resources and operators into virtual operational centres, in which all critical information is exposed to all stakeholders.

"This visible-from-anywhere information allows authorized users to provide the correct input, management and dollar accountability skills that will provide excellent comfort/energy performance with total accountability.

"A Web-based presentation of dynamic building information allows not only operators to operate from anywhere with full functionality, it allows interaction of contractors, equipment suppliers, and consultants to provide valuable feedback and feed-forward information to the building operating equation. Upper management can also participate by having browser-based bottom line screens that provide the dynamic proof of the success or failure of building performance.

"This new concentration and amplification of existing building operational resources and personnel will provide a strong re-focus on the values of good operating principles.

"This focus will cause management and operations to re-establish communications on what is important and what is required to achieve cost effective excellent building operations."

17.7 Summary

We are witnessing a large building automation industry converging with corporate enterprises. As technologies converge, clients' expectations are fueled by the ease of access and freedom of information on the Internet. The scope has increased to encompass environmental control, energy metering/accounting systems, lighting control, life safety/fire, security, communications, high tech tools, Web resources, and interactive information systems. With the trend towards IT-type solutions, video, card readers, and enterprise-based solutions are all becoming an active part of what is now called building automation.

17.8 Resources to Learn More

Below are some of the resources available free with the online magazine, AutomatedBuildings.com (http://www.automatedbuildings.com), that will help you learn more about the industry. The magazine is updated monthly and provides connection to almost all the dot coms in the evolving automated building industry.

Automated Building Systems Education
http://www.automatedbuildings.com/frame_education.htm

The rapid changing Automated Building industry has always required that all players be constantly re-educating themselves to keep current, but never has there been a time when this is so important.

Technology Roadmap for Intelligent Buildings
http://www.automatedbuildings.com/news/jan03/review/roadmap.htm

This 60-page glossy roadmap also available in a (66 pages) Adobe Acrobat PDF, 2.3 MB file is an excellent starting point for our large building automation industry to begin its reinvention and repackaging and to get on with the task of helping create an even better roadmap.

Intelligent Building Ranking System
This Task Force is developing an online tool intended to assist building owners/managers, the commercial real estate industry, and other industry stakeholders to assess the level of integrated systems within a building (a Building Intelligence Quotient - BIQ). A comprehensive list of intelligent building criteria has been developed as well as a "ranking matrix." The next phase of this project is to develop content detail on each line item in the matrix to be available as part of the online tool for ease of use by all industry stakeholders.

Middleware White Paper
The paper defines middleware and describes a number of case studies where middleware has provided a solution to integrate new intelligent building technology implementations with legacy systems.

Building Control Network Protocols White Paper
Communications protocols are simply a means by which different systems may communicate. They are the message formats and procedures used to transfer information, in an understandable form, from one device, or array of devices, to another. They permit products from different vendors to communicate with each other and interact to produce intelligent integrated building systems and manage and interface with these products as if the same vendor supplied them all. This paper, prepared by the Continental Automated Building Association (CABA) Intelligent & Integrated Buildings Council Building Protocol Task Group explores four of the most common protocols used today and compares a number of the parameters that "assist the large building industry to understand the strengths and overall features of the building control communication protocols that are available for use in designing and implementing an "Intelligent Building." CABA Resources can be found at: http://www.caba.org/councils/council-pubs.html

Best-Practices Guide for Evaluating IBT
This guide, authored by Kenneth P. Wacks, Ph.D., builds upon the Technology Roadmap for Intelligent Building Technology. This paper consists of criteria by which intelligent building technologies can be evaluated. The topics in this guide are important for various audiences, such as building owners and managers, intelligent building designers, and installers.

The Builconn Story
http://www.automatedbuildings.com/news/mar03/articles/builconn/bcon.htm

BuilConn intends to help integrators deliver a pragmatic approach to integration, not product or technology based, but solutions oriented.

Bring the Power of the Internet
http://www.automatedbuildings.com/news/jul00/articles/tridium/trid.htm

Perhaps no other technology has had the profound impact on virtually everything we do than the Internet**.** It is changing the way we make purchases, the way transactions are conducted between businesses, the way in which we obtain information, and the way we communicate with one another. The above article discusses how the power of the Internet is being adapted to real-time control and automation systems.

About the Author

Ken Sinclair is owner and editor of AutomatedBuildings.com, an online magazine and Web resource providing news and Web connection to the rapidly evolving industry that automates and implements intelligent, integrated buildings. He is a founding member and a past president of both the local chapter of Association of Energy Engineers (AEE) and the Vancouver Island Chapter of the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE). He has overseen 100+ computerized control conversions, working closely with building operators to insure acceptance of this new technology.